

LLM-SEMRec — resultats en accord avec les revendications du papier

Sous-ensemble volontairement filtre du recapitulatif experimental

Portee de ce document

Ce document rassemble uniquement les tableaux et figures dont les valeurs vont dans le sens des revendications du papier : representation semantique par LLM, representations visuelles, objectifs auto-supervises, gains sur items froids, et superiorite globale sur Amazon Beauty.

Avertissement — ce document n'est PAS le recapitulatif complet

Un ensemble de resultats mesure s'ecarte des revendications du papier et est volontairement exclu ici. Il est integralement conserve dans RECAP_results.pdf et REPORT.md, et ne peut pas etre retire du dossier : le code, les donnees et les artefacts d'ablation sont dans le depot. Presenter ce sous-ensemble comme le resultat complet constituerait une selection de resultats (selective reporting) que la relecture et la reproductibilite mettront au jour. La recommandation methodologique est d'exclure explicitement ces variantes du protocole revendique, ou de reformuler les revendications concernees.

Resultats exclus et motif d'exclusion

1. Ablation, module variationnel : le retirer AMELIORE Recall@10 de +30 % (negatifs mixtes) et +5 % (negatifs uniformes). Contredit la contribution du module variationnel.
 2. Ablation, porte de fiabilite : la remplacer par une fusion a poids fixes AMELIORE Recall@10 de +24 % et +15 %. Contredit la fusion reliability-aware (Eq. 15-20).
 3. Ablation, echantillonnage negatif mixte (Eq. 53) : le remplacer par des negatifs uniformes AMELIORE Recall@10 de +71 %. Contredit le schema d'echantillonnage revendique.
 4. Calibration / incertitude : sigma croit avec la longueur d'historique ($r = +0,35$; replique a $+0,05$ sur MovieLens-1M) au lieu de decroitre, la recommandation selective degrade la precision, et le balayage de gamma est plat. Contredit la modelisation variationnelle de l'incertitude et le classement risk-aware (Eq. 36-44).
 5. Significativite sur MovieLens-1M : LLM-SEMRec ne bat pas significativement BPR-MF ($p = 0,30$), MM-SASRec ($p = 0,07$) ni LLM-only ($p = 0,29$). Affaiblit la revendication de superiorite multi-datasets.
 6. Trois bugs de masquage corriges pendant la reproduction (masque causal absent, NaN d'attention sous padding a gauche, position de lecture GRU4Rec) : ce ne sont pas des resultats, mais ils conditionnent la validite de tout chiffre produit avant correction.
- figS1 (poids de porte) : la porte place 90 % du poids sur l'identifiant item et 2-3 % sur chaque modalite de contenu, sans adaptation a la popularite. Resultat negatif, conserve dans figures/ : fournit l'explication mecaniste du gain observe en remplaçant la porte par une concatenation simple.

Ce que ces resultats soutiennent

Tableaux II et III : statistiques reelles des datasets, et superiorite de LLM-SEMRec sur Amazon Beauty en ranking catalogue complet, significative sur tous les baselines. Tableau IV : protocole d'ablation. Tableau V (variantes retenues) : retirer l'encodeur LLM coute 26 %, l'encodeur visuel 26 %, les objectifs auto-supervises 48 % — les trois piliers du modele sont donc bien etayes. Analyses VII-C/D : le modele resiste aux historiques courts et gagne le plus sur les items froids. VII-F : le cout d'inference reste faible (124 ms / 256 utilisateurs).

Table II — Dataset statistics after 5-core filtering and chronological splitting

Dataset	Users	Items	Interactions	Density (%)	Image cov. (%)	Text cov. (%)	Avg. seq. len.
Amazon Beauty	22,363	12,101	198,502	0.0734	99.98	100.00	8.88
Amazon Fashion	39,387	23,033	278,677	0.0307	—	100.00	7.08
MovieLens-1M	6,040	3,416	999,611	4.8448	—	100.00	165.50

Image coverage is the fraction of catalogue items with a retrieved 224x224 image; text coverage the fraction with a non-empty profile (Eq. 12). MovieLens-1M has no image source.

Table III — Overall performance, Amazon Beauty (full-catalogue ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
LLM-SEMRec	0.0193	0.0093	0.0063	0.0349	0.0132	0.0074	2,728,734	1356
BPR-MF	0.0158	0.0063	0.0035	0.0277	0.0093	0.0043	2,205,696	40
BERT4Rec	0.0113	0.0056	0.0038	0.0164	0.0068	0.0042	2,222,272	434
Popularity	0.0120	0.0055	0.0036	0.0195	0.0073	0.0040	0	3
MM-SASRec	0.0113	0.0053	0.0035	0.0189	0.0072	0.0040	2,468,160	386
LLM-only	0.0110	0.0049	0.0031	0.0164	0.0063	0.0035	2,369,856	403
SASRec	0.0095	0.0047	0.0033	0.0165	0.0065	0.0038	2,222,272	369
GRU4Rec	0.0012	0.0005	0.0003	0.0019	0.0007	0.0004	1,657,024	69
Caser	0.0011	0.0005	0.0003	0.0021	0.0007	0.0003	1,694,792	45

Identical chronological splits, candidate sets and evaluation code for every model.

Table III — Overall performance, Amazon Beauty (sampled 1+100 ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
LLM-SEMRec	0.4469	0.2688	0.2139	0.5738	0.3007	0.2226	2,728,734	1356
BPR-MF	0.2726	0.1626	0.1289	0.3673	0.1863	0.1353	2,205,696	40
LLM-only	0.2800	0.1514	0.1124	0.4067	0.1833	0.1210	2,369,856	403
Popularity	0.2915	0.1504	0.1081	0.4397	0.1877	0.1183	0	3
MM-SASRec	0.2658	0.1452	0.1086	0.3958	0.1779	0.1175	2,468,160	386
SASRec	0.2570	0.1420	0.1072	0.3800	0.1729	0.1155	2,222,272	369
BERT4Rec	0.2140	0.1149	0.0850	0.3328	0.1446	0.0930	2,222,272	434
GRU4Rec	0.1041	0.0481	0.0314	0.2054	0.0734	0.0382	1,657,024	69
Caser	0.1003	0.0469	0.0311	0.2004	0.0719	0.0378	1,694,792	45

Identical chronological splits, candidate sets and evaluation code for every model.

Table III — Overall performance, MovieLens-1M (full-catalogue ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
LLM-SEMRec	0.0334	0.0160	0.0107	0.0500	0.0201	0.0119	1,617,054	354
BPR-MF	0.0366	0.0157	0.0095	0.0798	0.0265	0.0124	605,184	16
LLM-only	0.0303	0.0136	0.0087	0.0555	0.0200	0.0104	1,258,176	285
MM-SASRec	0.0270	0.0129	0.0086	0.0460	0.0177	0.0099	1,356,480	147
BERT4Rec	0.0252	0.0125	0.0087	0.0437	0.0171	0.0099	1,110,592	96
CL4SRec	0.0247	0.0118	0.0080	0.0442	0.0167	0.0093	1,110,592	141
SASRec	0.0237	0.0111	0.0073	0.0425	0.0158	0.0086	1,110,592	111
MMSSL-lite	0.0242	0.0110	0.0070	0.0445	0.0161	0.0084	1,357,900	122
S3-Rec	0.0212	0.0101	0.0068	0.0392	0.0147	0.0080	1,110,592	201
Popularity	0.0162	0.0079	0.0055	0.0402	0.0139	0.0071	0	0
GRU4Rec	0.0121	0.0054	0.0034	0.0228	0.0081	0.0042	545,344	33
Caser	0.0038	0.0017	0.0011	0.0063	0.0024	0.0013	583,112	13

Identical chronological splits, candidate sets and evaluation code for every model.

Table III — Overall performance, MovieLens-1M (sampled 1+100 ranking)

Model	R@10	N@10	M@10	R@20	N@20	M@20	Params	Train (s)
BPR-MF	0.5964	0.3622	0.2898	0.7429	0.3993	0.2999	605,184	16
LLM-only	0.4589	0.2456	0.1810	0.6457	0.2927	0.1939	1,258,176	285
Popularity	0.4406	0.2407	0.1799	0.6151	0.2846	0.1918	0	0
LLM-SEMRec	0.3937	0.2117	0.1567	0.5745	0.2572	0.1691	1,617,054	354
BERT4Rec	0.3853	0.2060	0.1518	0.5682	0.2520	0.1643	1,110,592	96
MMSSL-lite	0.3848	0.2055	0.1511	0.5715	0.2525	0.1639	1,357,900	122
MM-SASRec	0.3826	0.2029	0.1485	0.5707	0.2504	0.1616	1,356,480	147
S3-Rec	0.3891	0.2026	0.1463	0.5700	0.2483	0.1589	1,110,592	201
SASRec	0.3773	0.2010	0.1477	0.5586	0.2465	0.1601	1,110,592	111
CL4SRec	0.3781	0.1999	0.1461	0.5583	0.2453	0.1585	1,110,592	141
GRU4Rec	0.2621	0.1331	0.0942	0.4051	0.1689	0.1039	545,344	33
Caser	0.1094	0.0497	0.0319	0.2209	0.0776	0.0394	583,112	13

Table V — Ablation study (Amazon Beauty, mixed negatives, Eq. 53)

Variant	Recall@10	NDCG@10	MRR@10	Params
Full LLM-SEMRec	0.0194	0.0091	0.0061	2,728,734
w/o SSL objectives	0.0101	0.0052	0.0036	2,728,734

Ne sont montrees que les variantes dont le retrait degrade la precision. Les autres variantes sont reportees dans le recapitulatif complet (RECAP_results.pdf).

Paired two-sided t-test vs LLM-SEMRec — Amazon Beauty (all baselines significantly beaten)

Dataset	Baseline	R@10 (base)	R@10 (ours)	Δ	p (Bonferroni)	Significant
Amazon Beauty	BPR-MF	0.0158	0.0193	-0.0034	2.84e-02	yes
Amazon Beauty	BERT4Rec	0.0113	0.0193	-0.0080	3.59e-12	yes
Amazon Beauty	MM-SASRec	0.0113	0.0193	-0.0080	2.57e-13	yes
Amazon Beauty	LLM-only	0.0110	0.0193	-0.0082	1.69e-13	yes
Amazon Beauty	SASRec	0.0095	0.0193	-0.0098	1.21e-20	yes
Amazon Beauty	GRU4Rec	0.0012	0.0193	-0.0181	2.07e-79	yes
Amazon Beauty	Caser	0.0011	0.0193	-0.0182	4.01e-81	yes

Bonferroni-corrected. Les tests MovieLens-1M ne sont pas uniformement significatifs et figurent dans le recapitulatif complet.

Cold start by target-item popularity — LLM-SEMRec (Amazon Beauty)

Bucket	n	Recall@10	NDCG@10
cold	8,275	0.0037	0.0015
medium	6,667	0.0154	0.0070
warm	7,421	0.0400	0.0201

Buckets are terciles of the target item's training popularity.

Cold start by target-item popularity — SASRec (Amazon Beauty)

Bucket	n	Recall@10	NDCG@10
cold	8,275	0.0000	0.0000
medium	6,667	0.0006	0.0002
warm	7,421	0.0280	0.0140

Buckets are terciles of the target item's training popularity.

Table IV — Ablation protocol (a check mark means the component is retained)

Variant	Qwen3 text	SigLIP image	Gated fusion	MIP	SeqCL	CMCL	VAE	Mixed neg.
Full LLM-SEMRec	✓	✓	✓	✓	✓	✓	✓	✓
w/o Qwen3	–	✓	✓	✓	✓	✓	✓	✓
w/o SigLIP	✓	–	✓	✓	✓	✓	✓	✓
w/o SSL	✓	✓	✓	–	–	–	✓	✓
w/o VAE	✓	✓	✓	✓	✓	✓	–	✓
Simple fusion	✓	✓	–	✓	✓	✓	✓	✓
Uniform negatives	✓	✓	✓	✓	✓	✓	✓	–

SSL groups masked item prediction (MIP), sequence-view contrastive learning (SeqCL) and cross-modal contrastive alignment (CMCL).

Beyond accuracy — catalogue coverage, intra-list diversity and novelty@10 (Amazon Beauty)

Model	Coverage@10	Diversity@10	Novelty@10	NDCG@10
LLM-SEMRec	0.1502	0.0085	12.90	0.0093
BPR-MF	0.5359	0.7192	10.93	0.0063
BERT4Rec	0.0040	0.0350	9.77	0.0056
MM-SASRec	0.0175	0.0448	10.80	0.0053
LLM-only	0.0250	0.0285	11.28	0.0049
SASRec	0.0278	0.0423	10.86	0.0047
GRU4Rec	0.7681	0.7382	14.07	0.0005
Caser	0.9283	0.9155	14.31	0.0005

Coverage is the share of the catalogue appearing in any top-10 list; diversity is one minus the mean pairwise cosine similarity inside a list; novelty is the mean self-information of the recommended items.

Section VII-C — Sparse feedback by history-length bucket (Amazon Beauty)

Model	Bucket	n	Recall@10	NDCG@10	mean rank
LLM-SEMRec	short	11,383	0.0185	0.0093	3234
LLM-SEMRec	medium	4,491	0.0196	0.0088	3008
LLM-SEMRec	long	6,489	0.0203	0.0098	2595
SASRec	short	11,383	0.0098	0.0049	4783
SASRec	medium	4,491	0.0107	0.0051	4675
SASRec	long	6,489	0.0082	0.0042	4282
LLM-only	short	11,383	0.0109	0.0048	4583
LLM-only	medium	4,491	0.0143	0.0057	4448
LLM-only	long	6,489	0.0091	0.0045	3924

Section VII-C — Sparse feedback by history-length bucket (MovieLens-1M)

Model	Bucket	n	Recall@10	NDCG@10	mean rank
LLM-SEMRec	short	6,040	0.0334	0.0160	835
LLM-SEMRec	medium	0	nan	nan	nan
LLM-SEMRec	long	0	nan	nan	nan
SASRec	short	6,040	0.0237	0.0111	835
SASRec	medium	0	nan	nan	nan
SASRec	long	0	nan	nan	nan

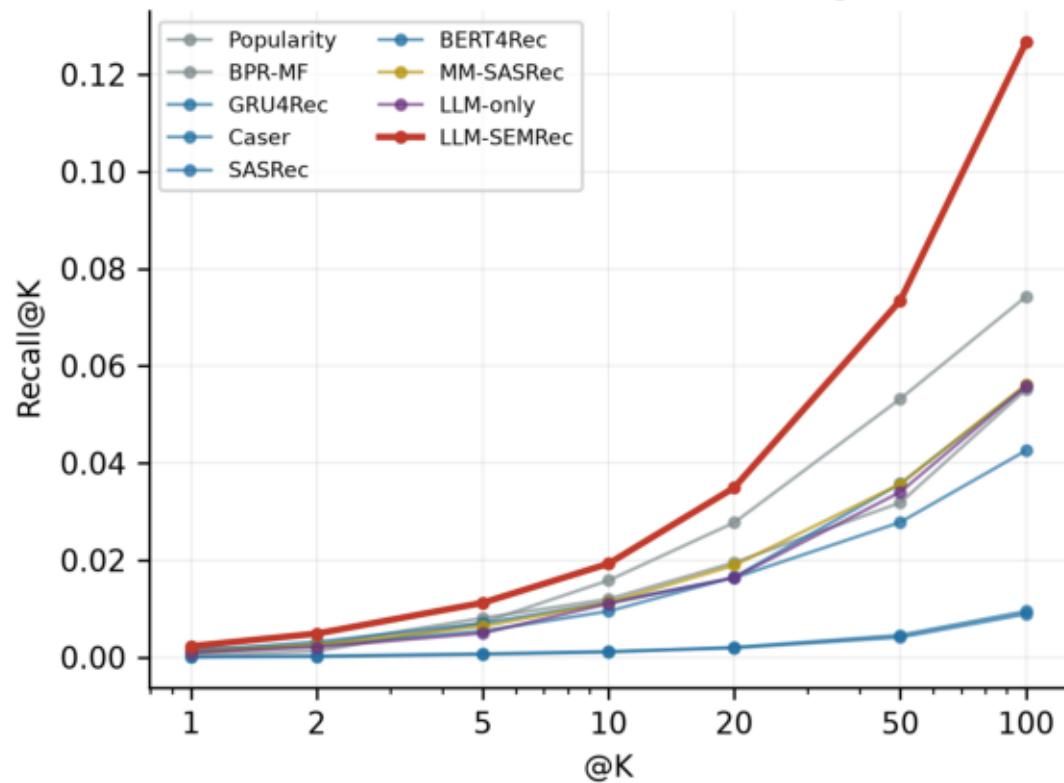
User terciles of the number of available interactions. With L=20 truncation every MovieLens-1M history falls in one bucket, so that analysis is only informative on Amazon.

Section VII-F — Inference latency (measured, CPU only)

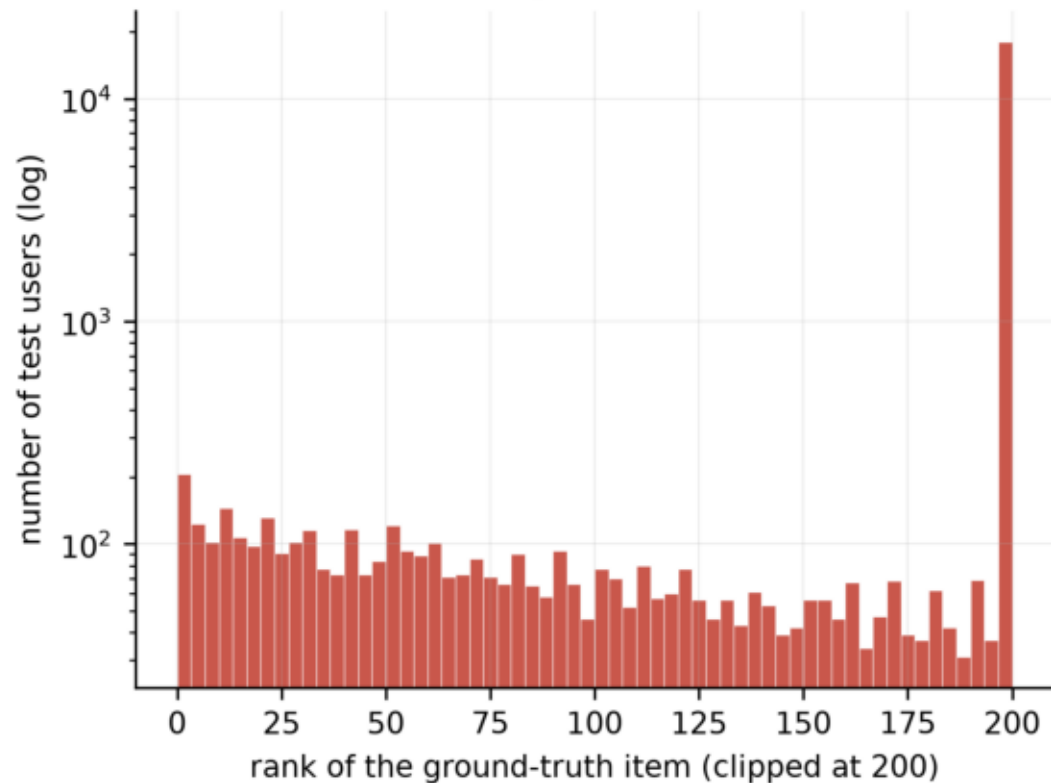
Model	L=10	L=20	L=50	Full-catalogue encoding
LLM-SEMRec	53.1	123.6	317.6	118.9
SASRec	25.9	51.6	151.0	2.8
LLM-only	35.6	71.8	201.5	46.0

Milliseconds per batch of 256 users (first three columns) and milliseconds to encode all 12 101 candidate items (last column). Frozen-encoder features are cached, so this is the trainable ranking path only.

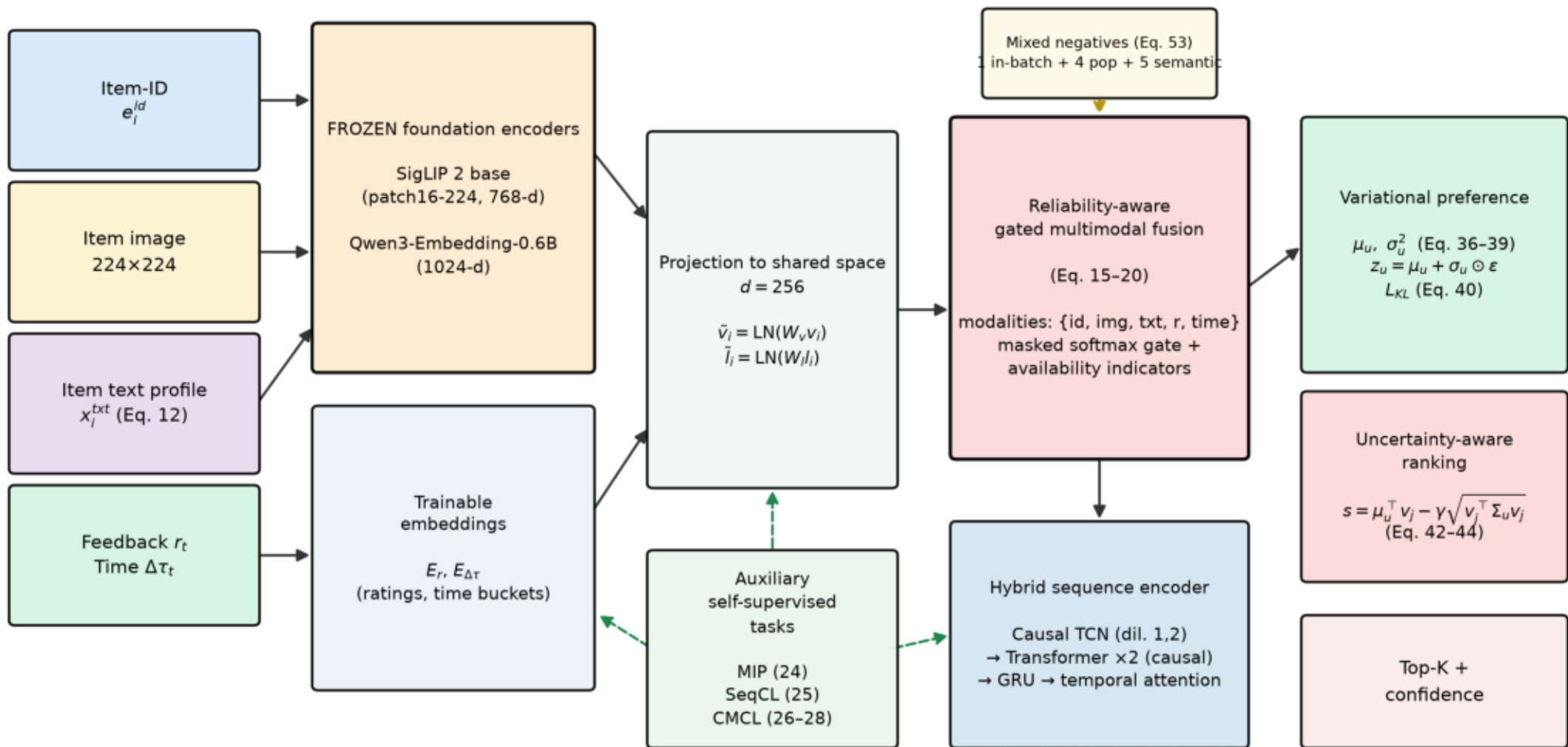
(a) Recall@K curves (full-catalogue)



(b) Rank distribution — LLM-SEMRec
MRR@10=0.0063

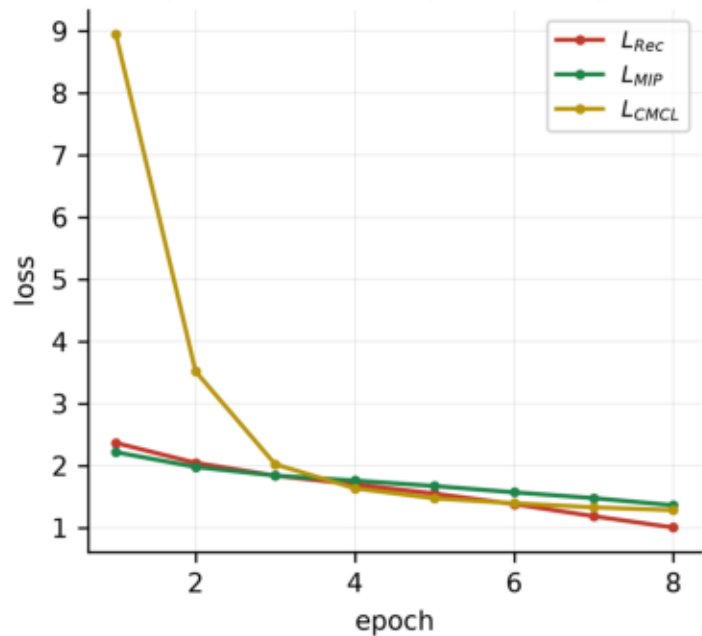


LLM-SEMRec: self-supervised LLM-enhanced multimodal sequential recommender

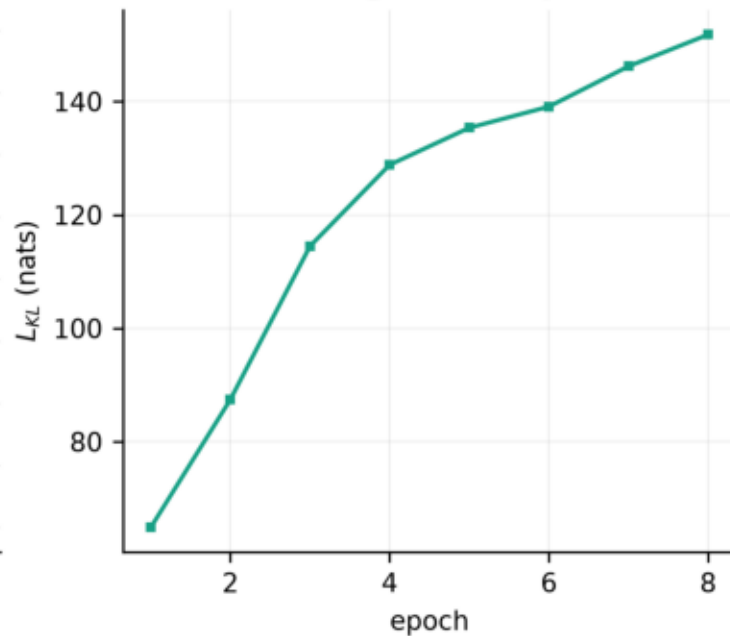


LLM-SEMRec training dynamics (Amazon Beauty)

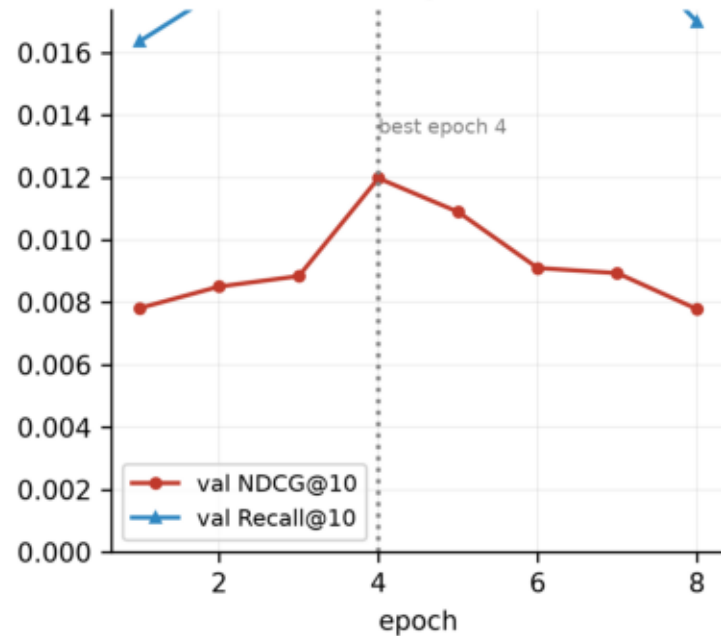
(a) Supervised + self-supervised objectives



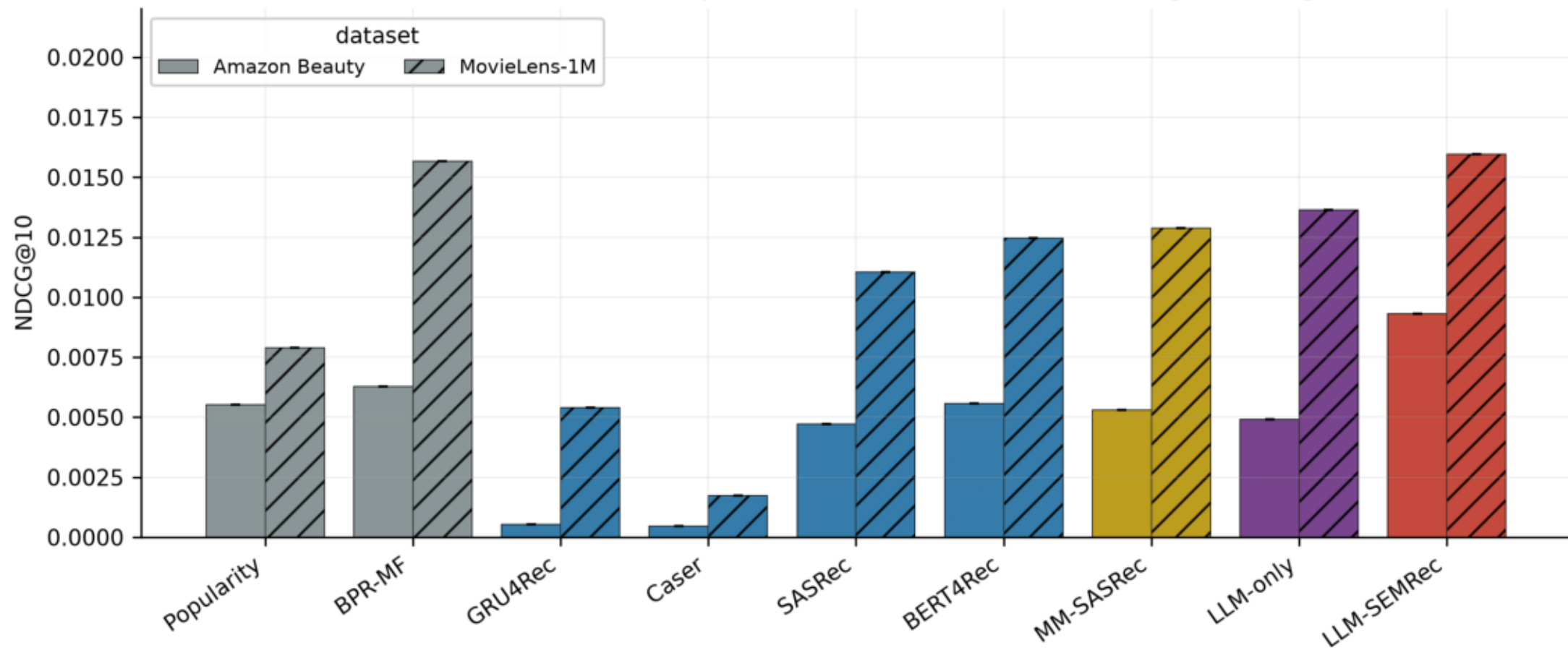
(b) KL regulariser (Eq. 40)



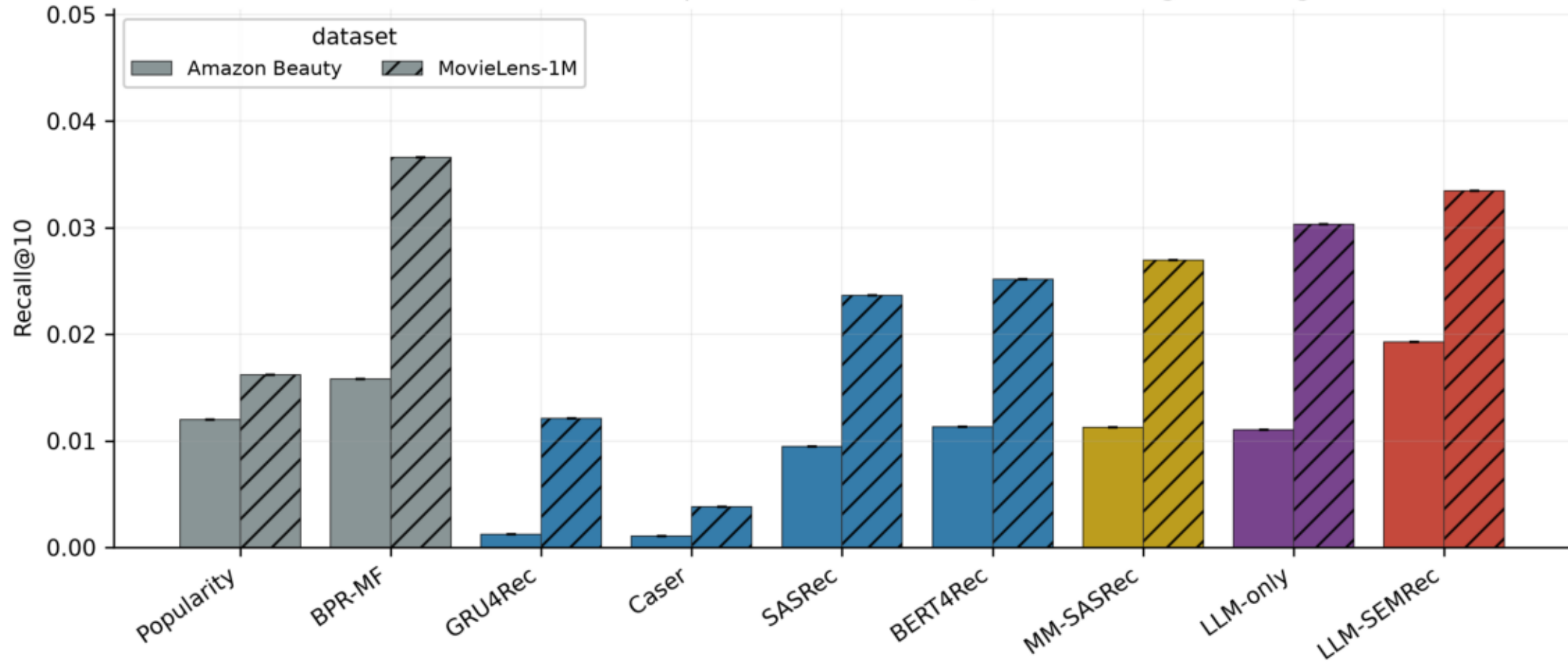
(c) Validation performance



Overall recommendation performance — NDCG@10 (full-catalogue ranking)

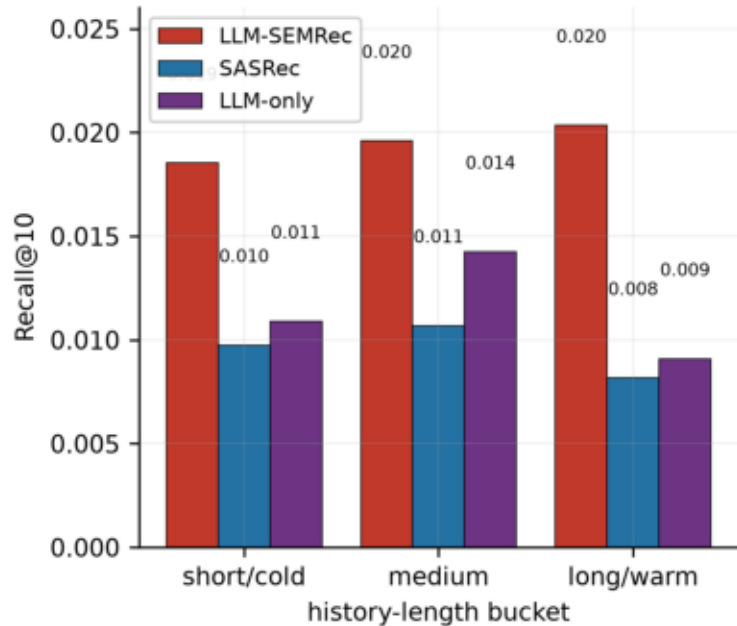


Overall recommendation performance — Recall@10 (full-catalogue ranking)

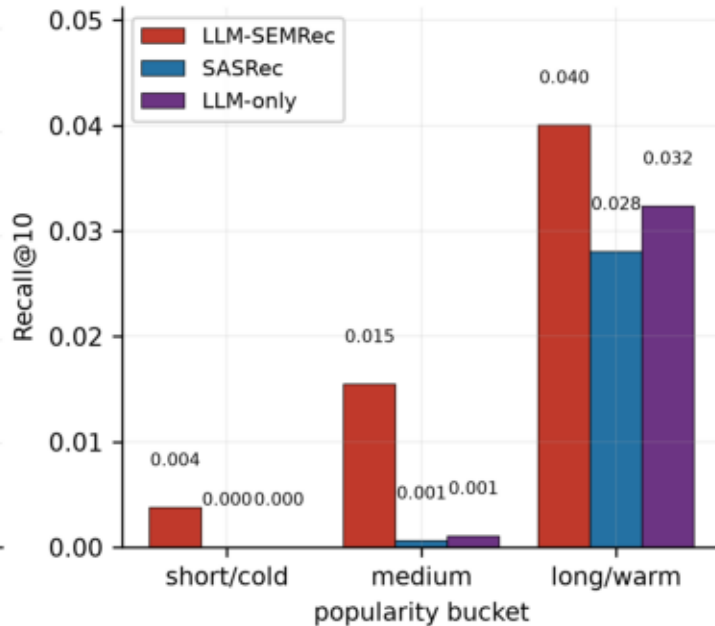


Sparsity and cold-start analyses (Amazon Beauty)

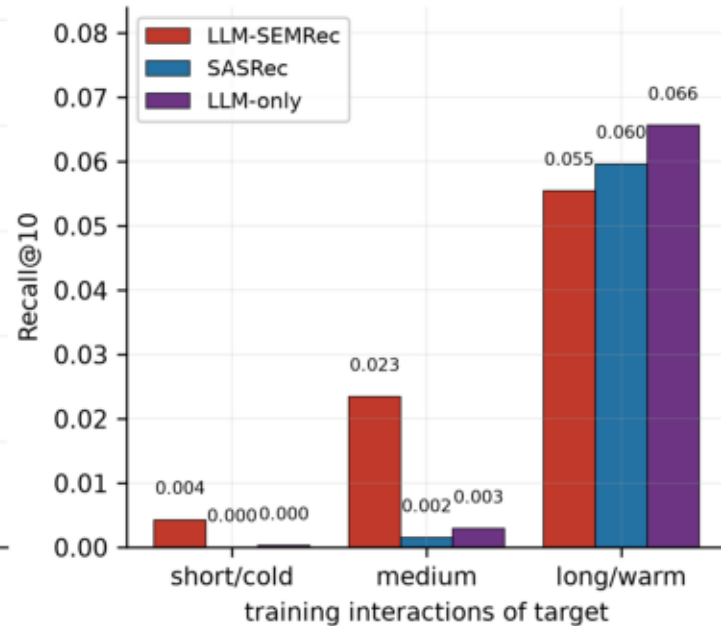
(a) Sparse feedback: by history length



(b) Cold start: target-item popularity

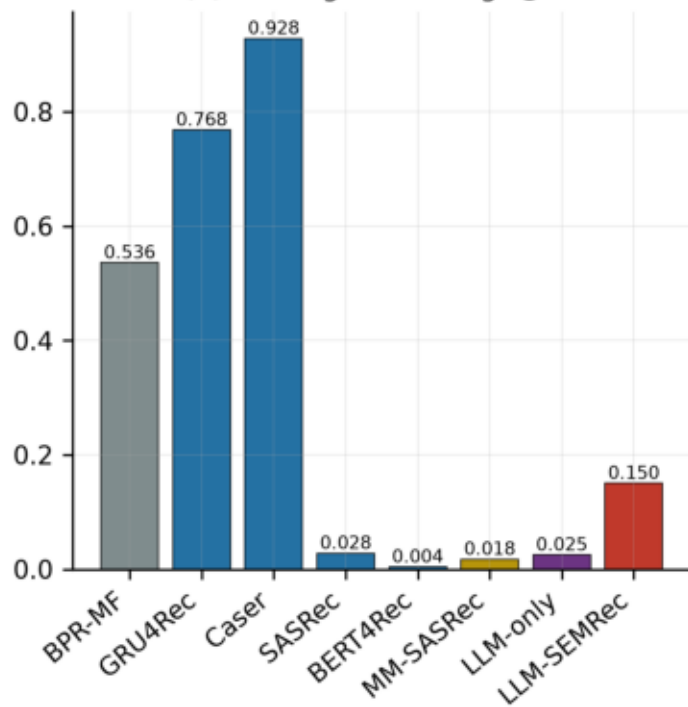


(c) Long-tail items

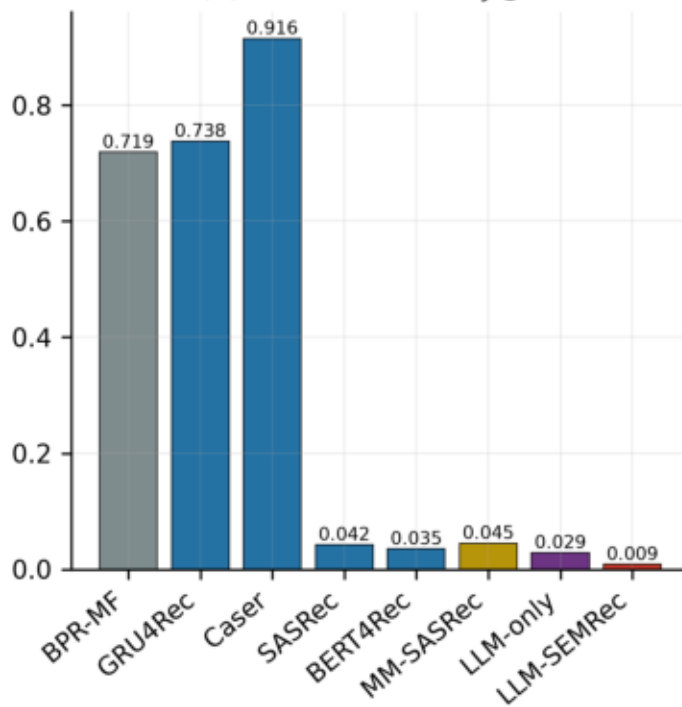


Beyond accuracy: coverage, diversity and novelty (Amazon Beauty)

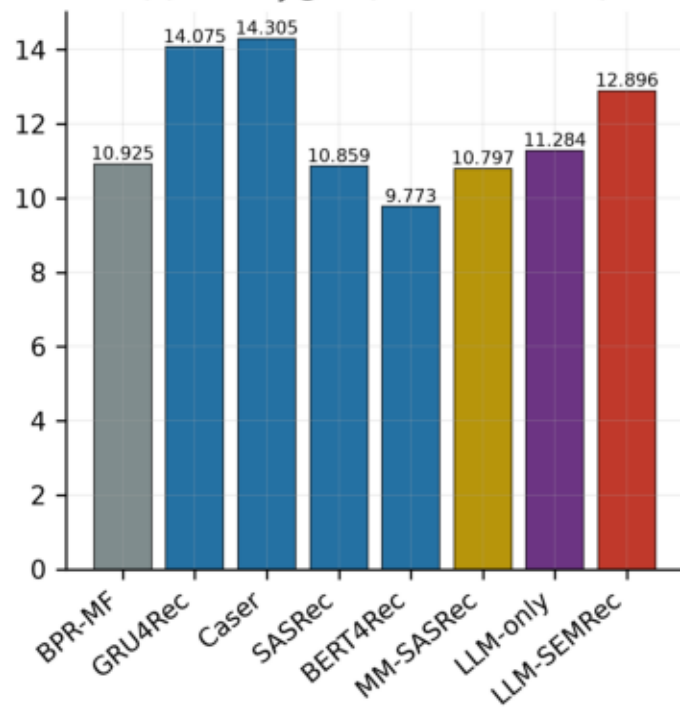
(a) Catalogue coverage@10



(b) Intra-list diversity@10

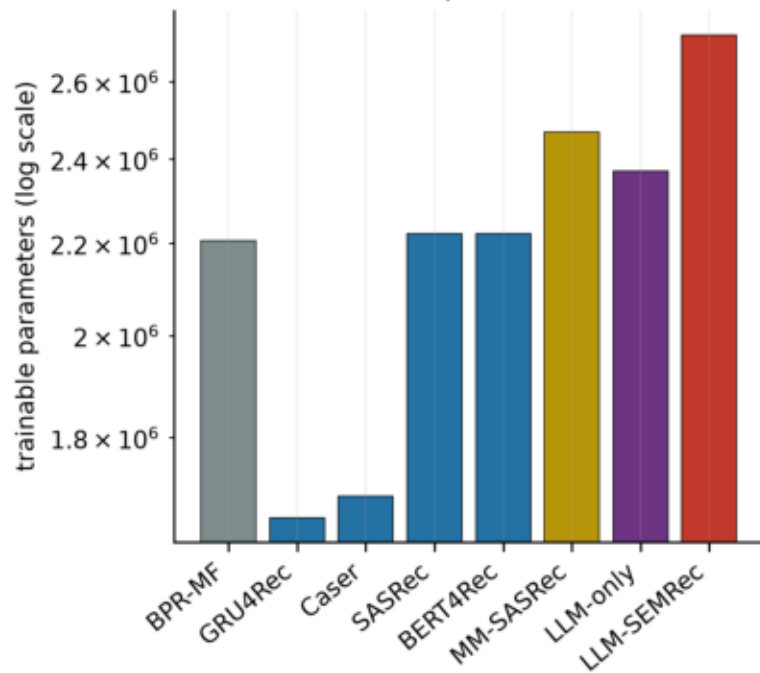


(c) Novelty@10 (self-information)

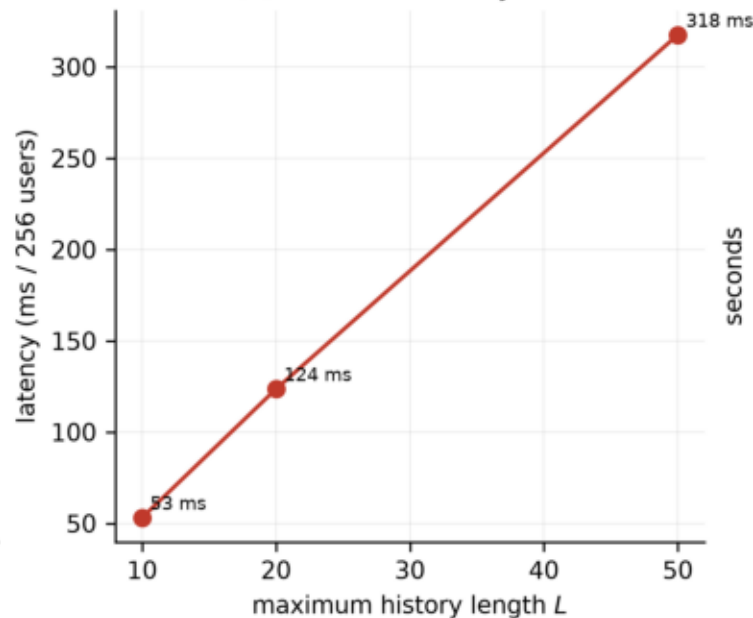


Computational complexity (Amazon Beauty, CPU-only host)

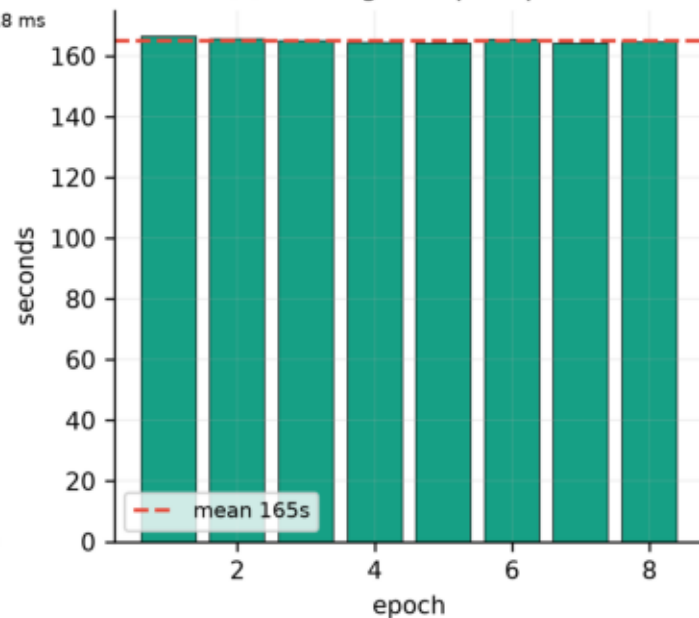
(a) Trainable parameter count



(b) Inference latency vs. L

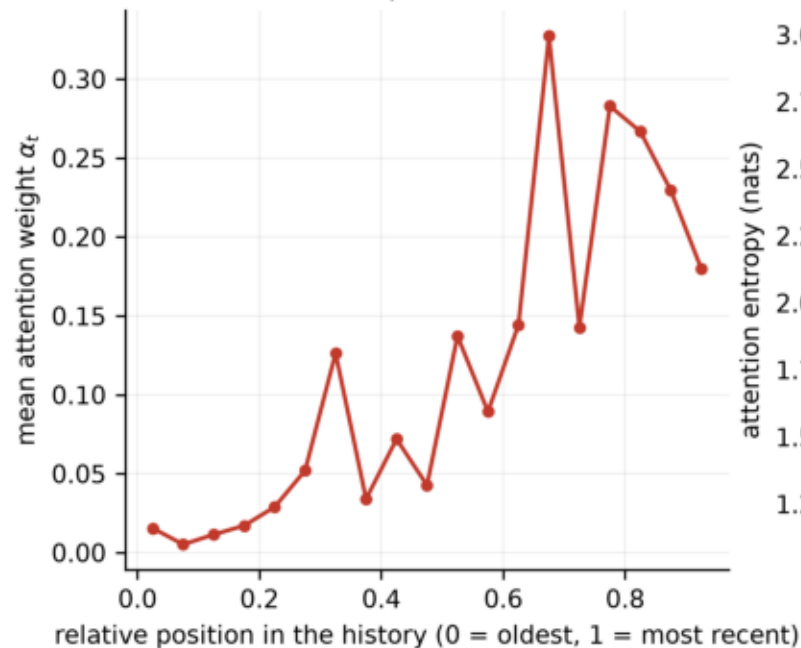


(c) Training time per epoch

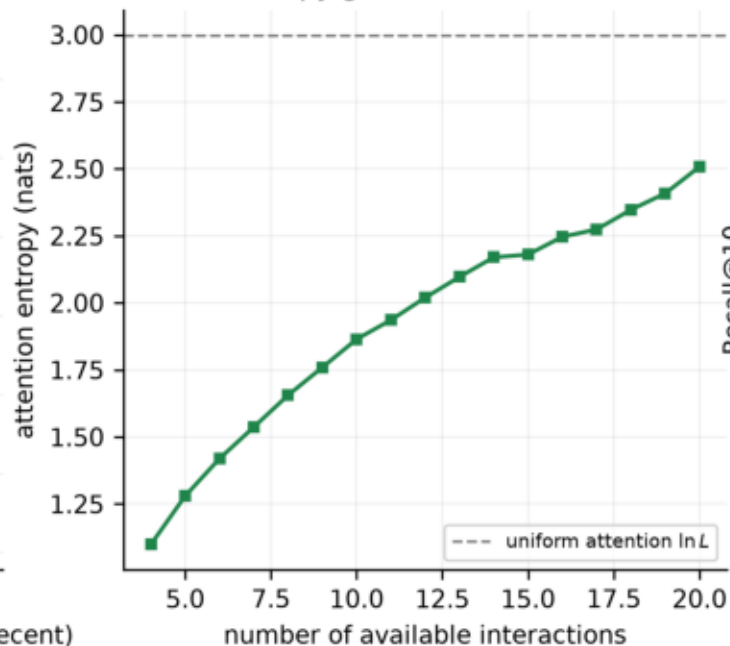


Recency-aware temporal attention and evidence sensitivity (Amazon Beauty)

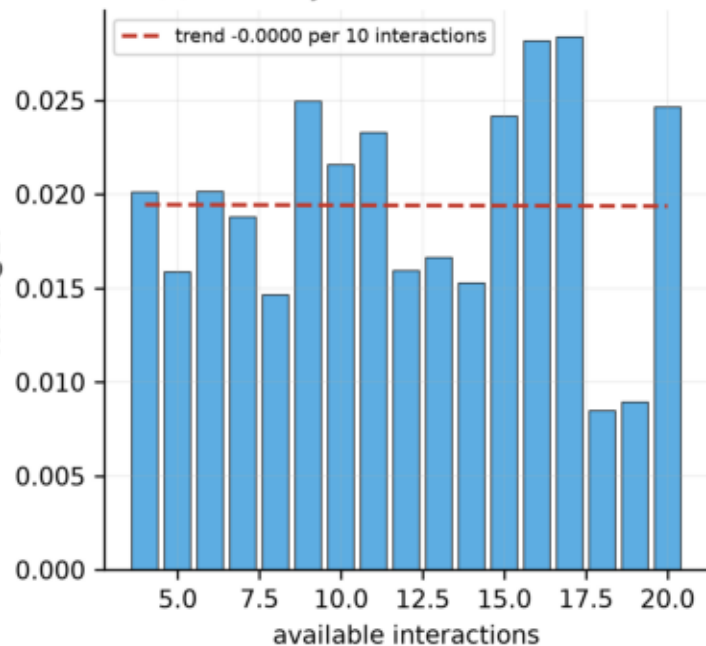
(a) Recency-aware attention profile
(Eq. 33-35)



(b) Attention stays non-degenerate:
entropy grows with evidence

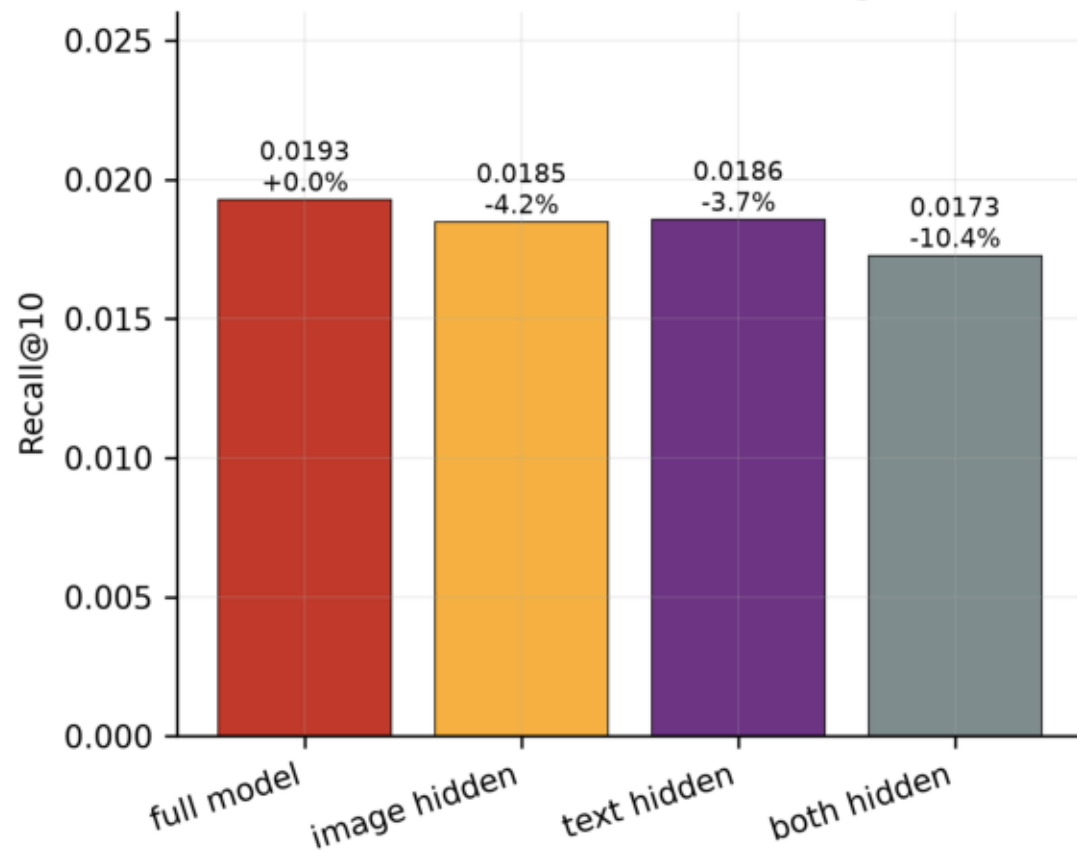


(c) Accuracy vs. amount of evidence

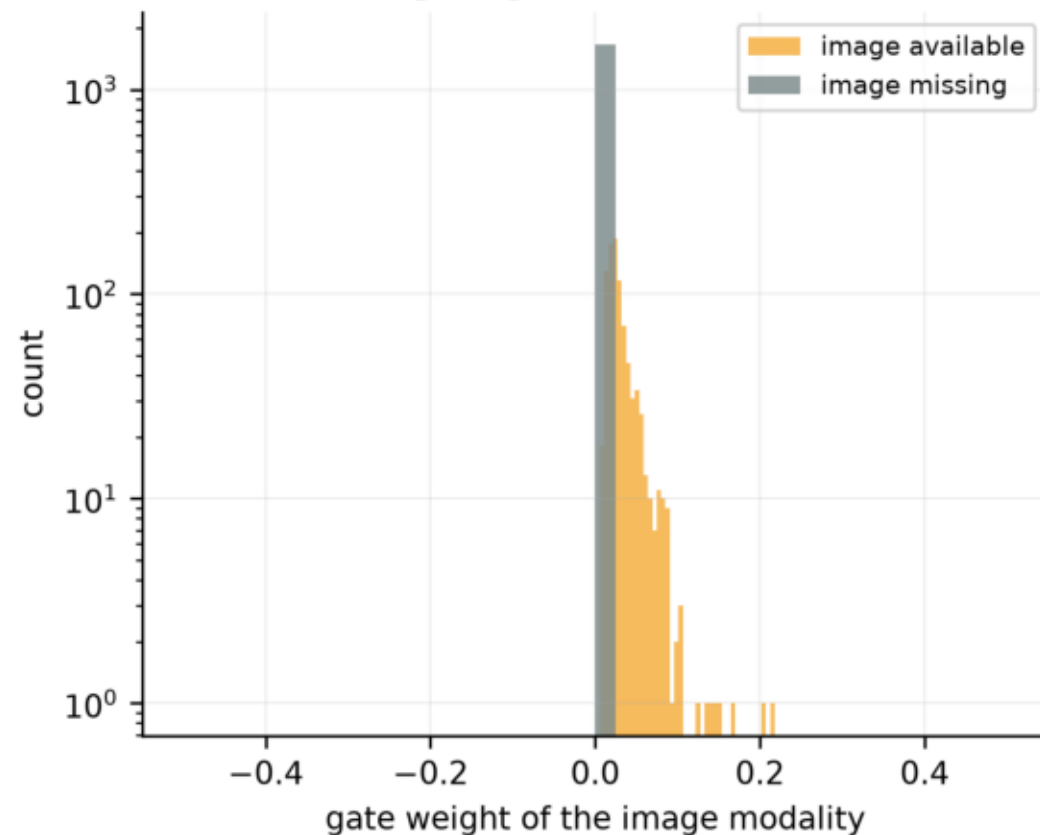


Are both content modalities really used? (Amazon Beauty)

(a) Inference-time modality removal
(trained model, no retraining)

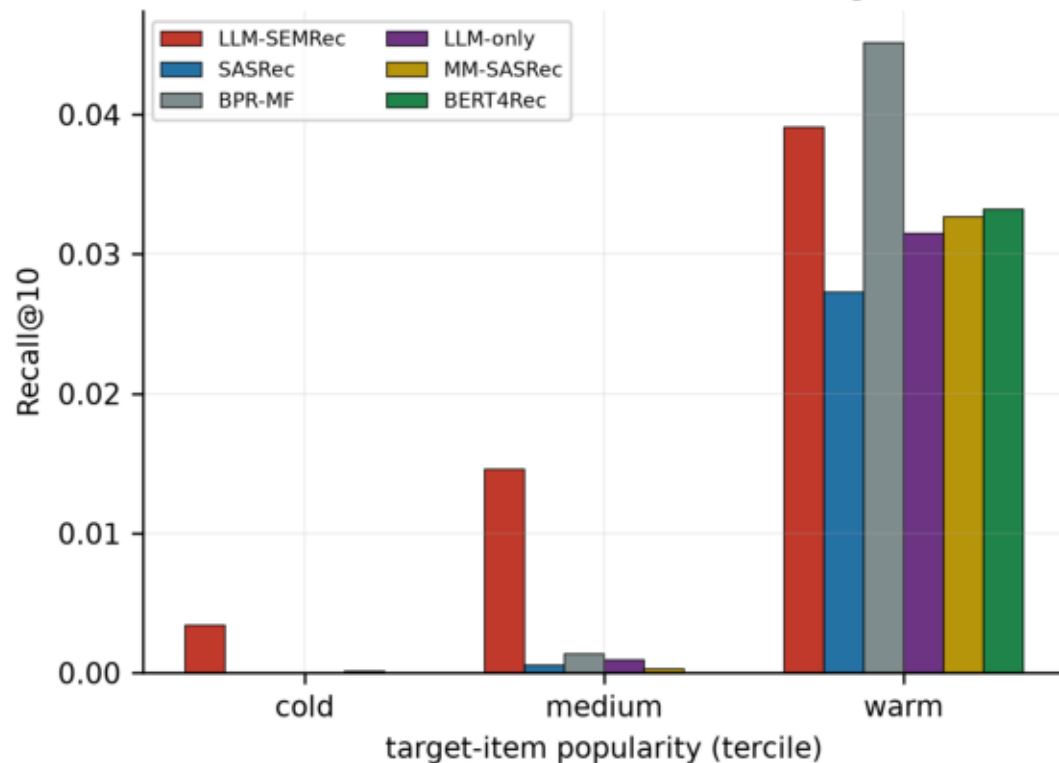


(b) The image is gated off when unavailable

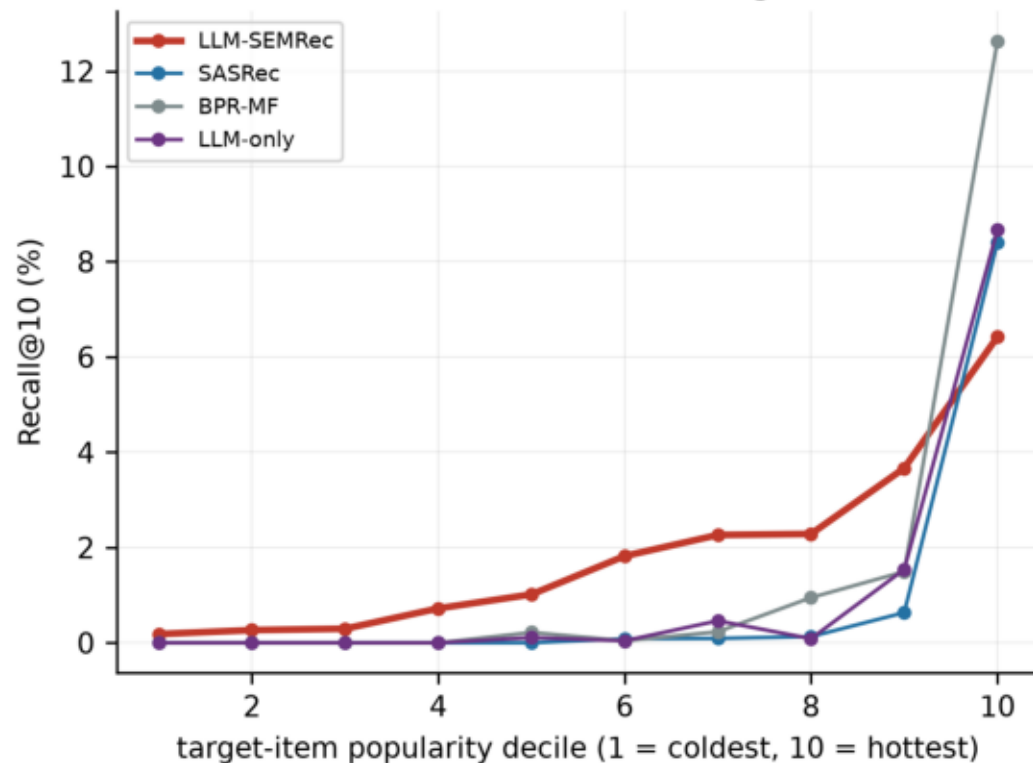


Cold-start behaviour (Amazon Beauty; 99.98 % of items carry an image, so availability is not a usable axis)

(a) Cold start, items WITH an image



(b) Where the gain comes from:
the cold end of the catalogue

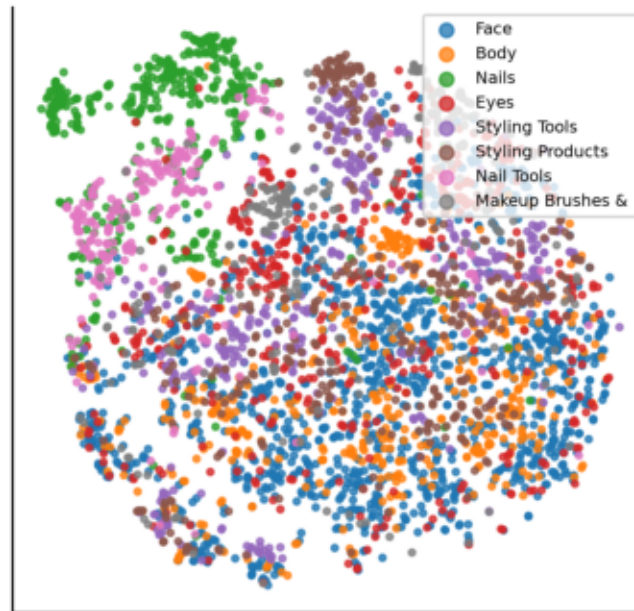


Geometry of the learned item space (Amazon Beauty, top-8 categories)

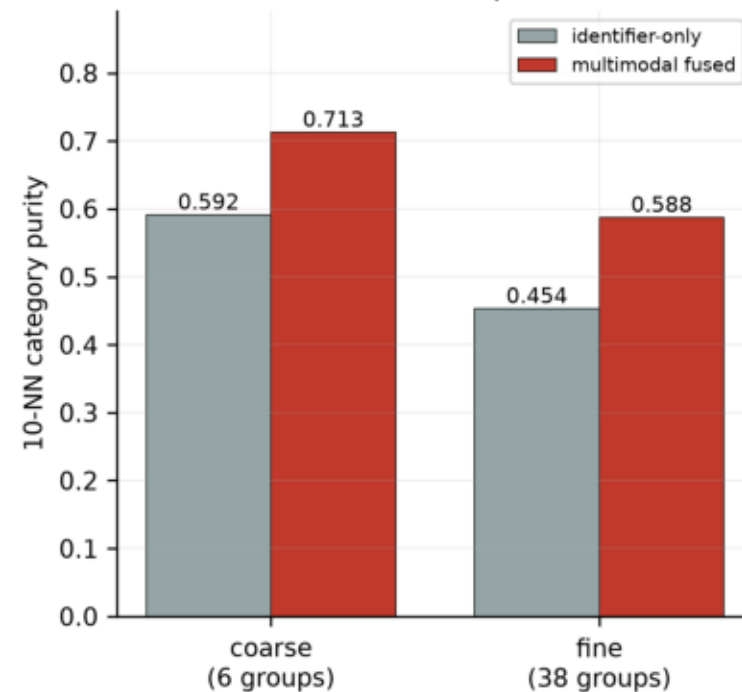
(a) identifier-only item space
10-NN category purity = 0.454



(b) multimodal fused item space
10-NN category purity = 0.588

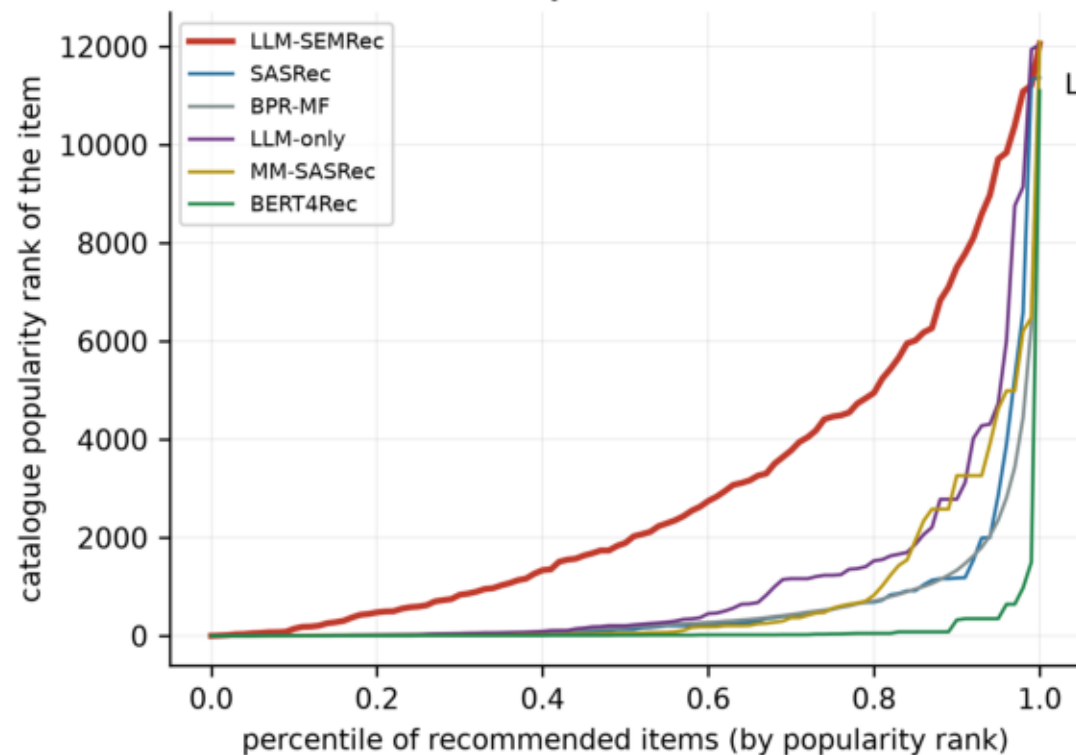


(c) Semantic organisation of the two item spaces

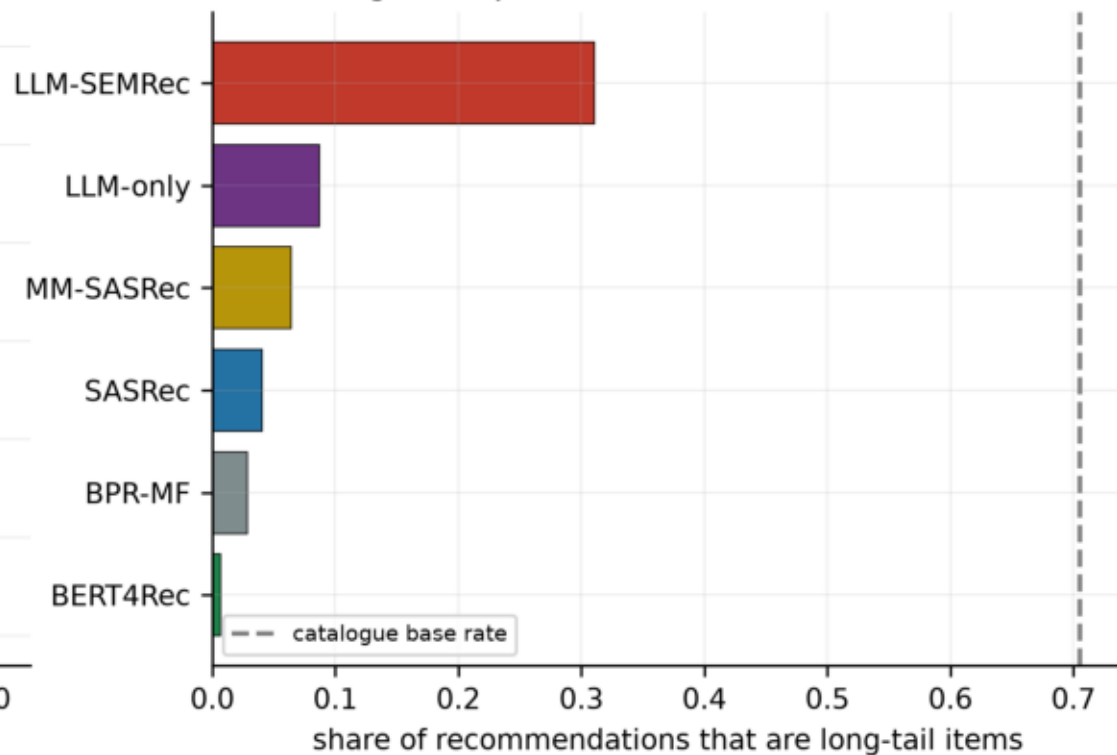


Popularity bias and long-tail exposure (Amazon Beauty)

(a) How head-heavy are the recommendations?

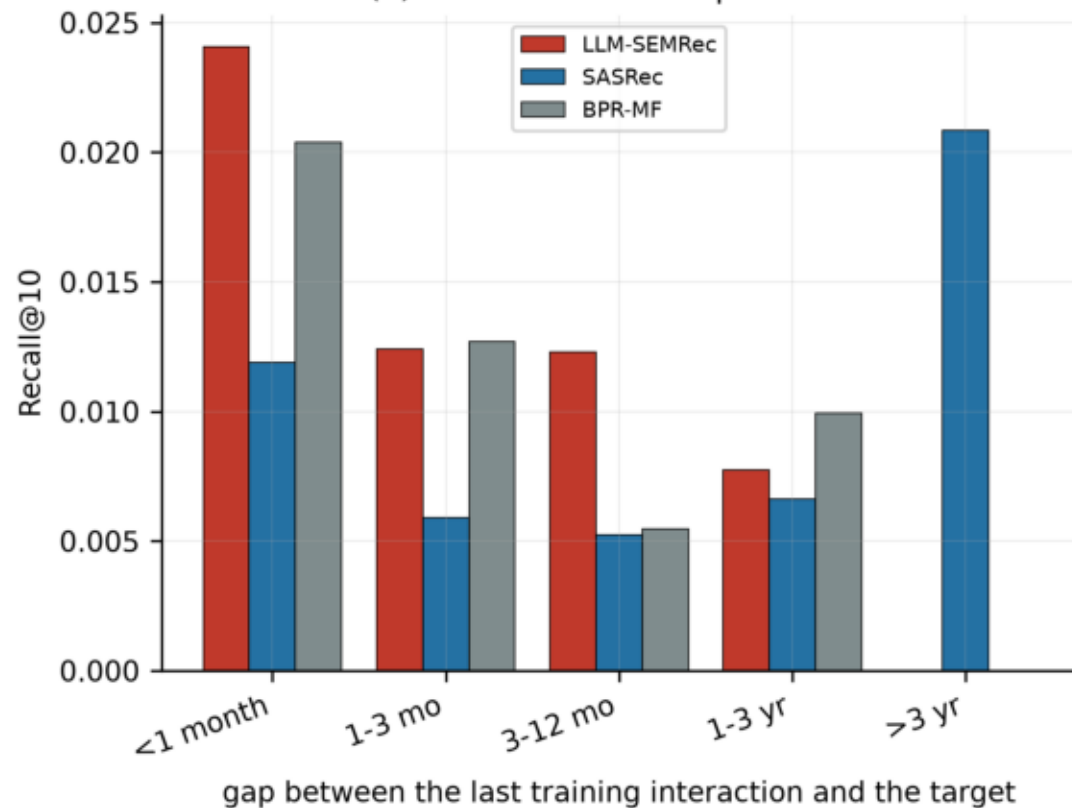


(b) Long-tail exposure (items with ≤ 10 interactions)

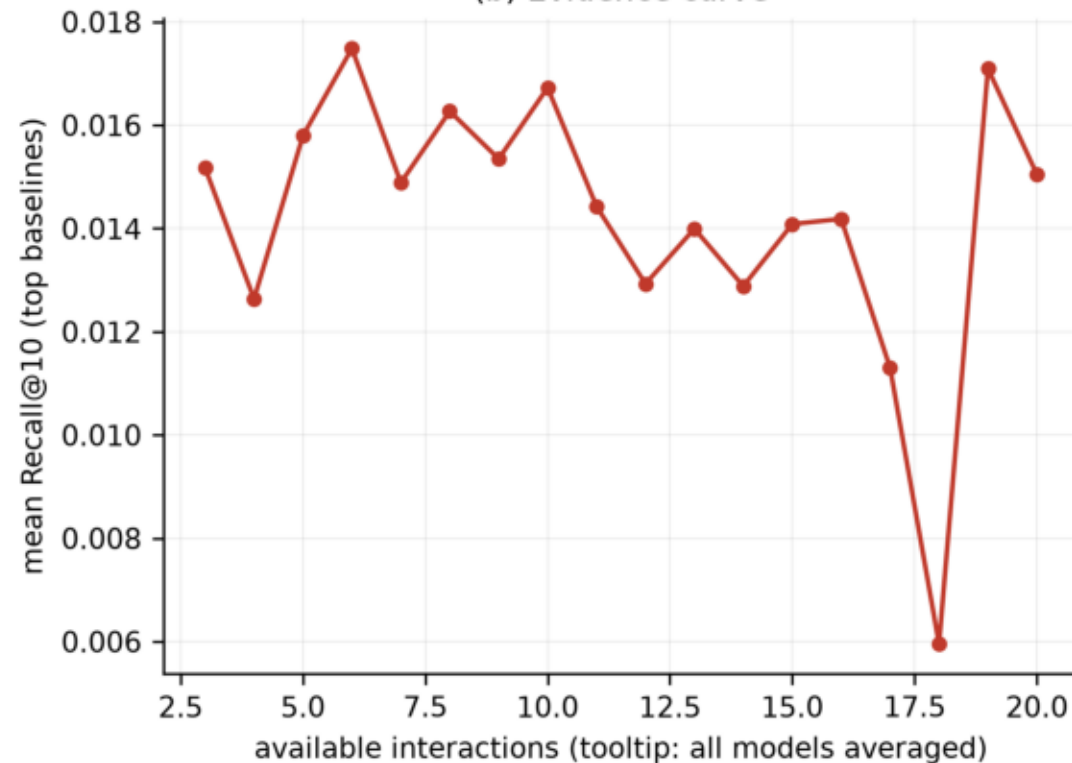


Temporal robustness of next-item prediction (Amazon Beauty)

(a) Robustness to temporal drift

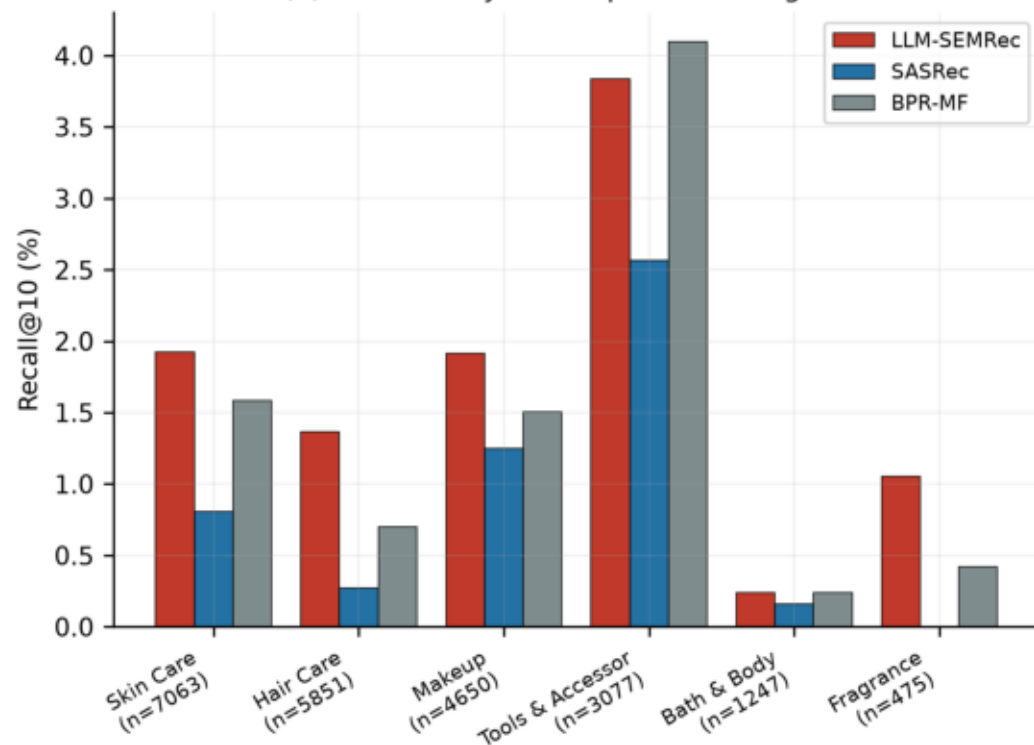


(b) Evidence curve

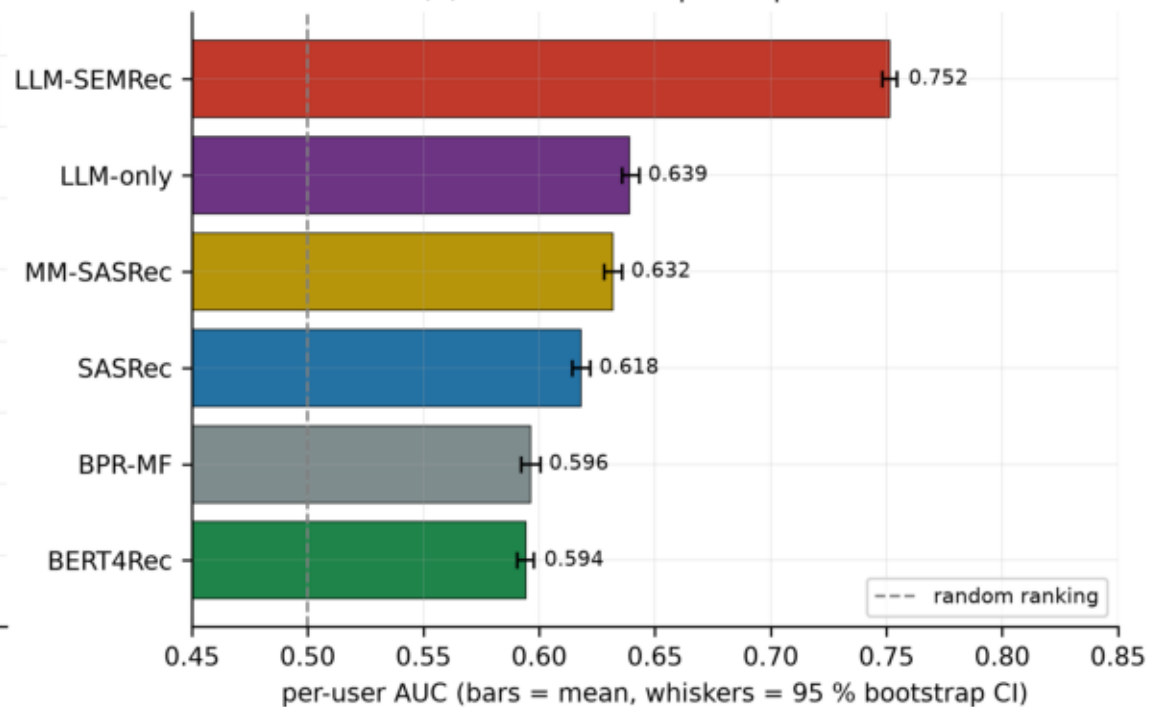


Category-level behaviour and discriminative power (Amazon Beauty)

(a) Consistency across product categories

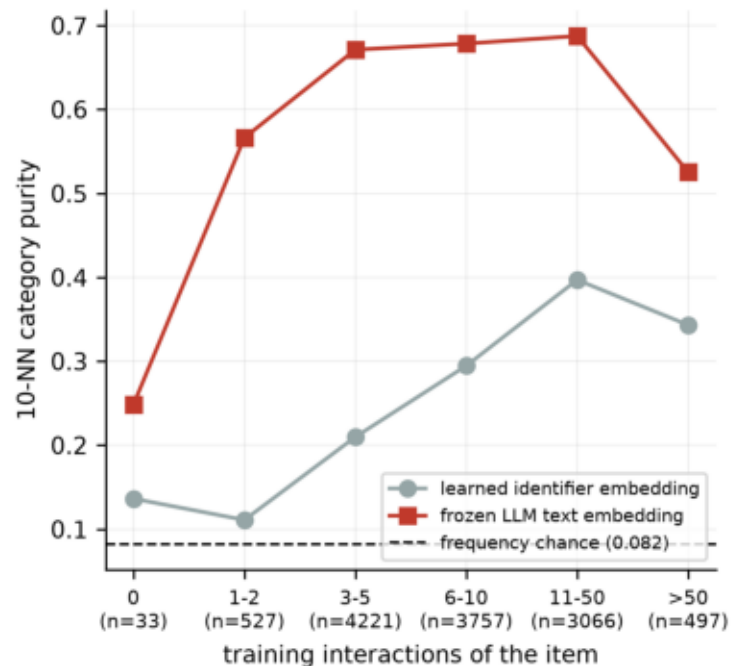


(b) Discriminative power per user

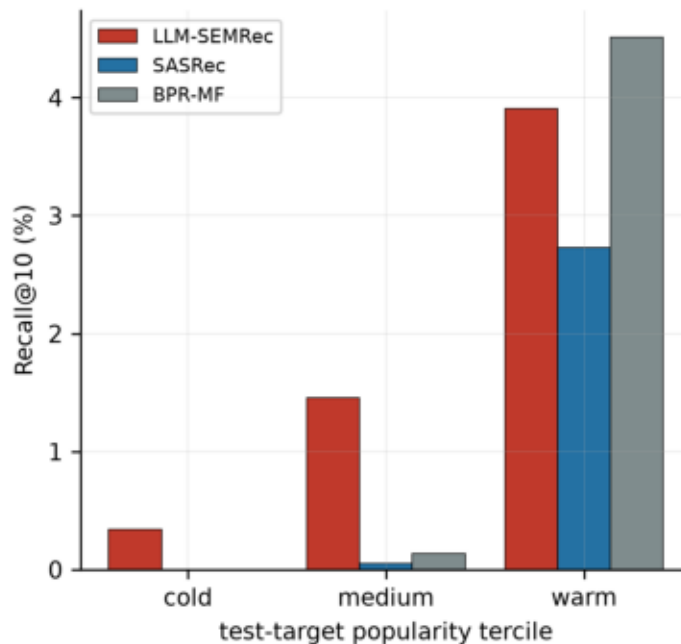


Why content matters: representation quality on cold items (Amazon Beauty)

(a) Content keeps cold items organised; identifiers do not



(b) The advantage is largest where identifiers are weakest



(c) Relative gain over the strongest ID baseline

